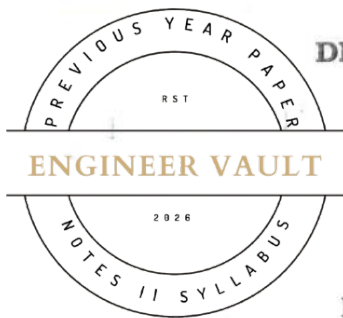




**DR. BABASAHEB AMBEDKAR MARATHWADA UNIVERSITY,
CHHATRAPATI SAMBHAJINAGAR.**



NAAC- 'A⁺' Grade

CIRCULAR NO.SU/Engg./College/NEP/83/2025

It is hereby inform to all concerned that, the syllabi prepared by the Board of Studies and recommended by the Dean, Faculty of Science & Technology, **Academic Council at its meeting held on 09 May 2025 has been accepted** the following Syllabi and Second Year Open Elective Basket for B. E. & B. Tech. under the Faculty of Science & Technology as per Norms of National Education Policy – 2020 as appended herewith.

Sr.No.	Syllabi
1.	Second Year B. E./B. Tech. Civil Engineering
2.	Second Year B.E./B.Tech. Mechanical Engineering
3.	Second Year B. E./B. Tech. Information Technology
4.	Second Year B. E./B. Tech. Electronics & Telecommunication Engineering/ Electronics Engineering / Electronics & Communication Engineering.
5.	Second Year B. E./B. Tech. Electrical and Electronics Engineering/ Electrical Engineering/ Electrical, Electronics and Power
6.	Second Year B. E./B. Tech. Computer Science and Engineering/ Computer Science/Computer Engineering
7.	Second Year B. E./B. Tech. Artificial Intelligence/ Artificial Intelligence and Machine Learning / Artificial Intelligence and Data Science

This is effective from the Academic Year 2025-26 and onwards.

All concerned are requested to note the contents of this circular and bring the notice to the students, teachers and staff for their information and necessary action.

University Campus,
Chhatrapati Sambhajinagar
- 431 004.

REF.No.SU/NEP/2025/ 856-6)

Date:- 29/ 05/ 2025.

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**Deputy Registrar,
Syllabus Section**

Copy forwarded and necessary action to :-

- 1] **The Principal of all Affiliated Colleges,** Dr. Babasaheb Ambedkar Marathwada University
- 2] The Director, University Network & Information Centre, UNIC, Dr.Babasaheb Ambedkar Marathwada University with a request to upload this Circular on University Website
- 3] The Director, Board of Examinations & Evaluation, Dr. Babasaheb Ambedkar Marathwada University, Chhatrapati Sambhajinagar.

Copy to :-

- 1] PA to the Hon'ble Vice-Chancellor,
- 2] PA to the Pro. Vice-Chancellor,
- 3] PA to the Registrar,
Dr. Babasaheb Ambedkar Marathwada University, Chhatrapati Sambhajinagar.

**Dr. Babasaheb Ambedkar Marathwada University
Chhatrapati Sambhajinagar- 431001**



**Four Year UG Engineering/Technology
Programme (B.E. /B. Tech)**

**Structure of SY B.E./B.Tech Programme
(Mechanical Engineering)**

(AS PER NEP-2020)

Effective from 2025-26

Prof. Dr. Ashok Keche

FACULTY OF SCIENCE AND TECHNOLOGY

B. E. / B. Tech (Mechanical Engineering)

Syllabus Structure w.e.f. 2025-26

(As Per NEP-2020)

Semester-III

Option 1 : Student can choose this option (Option 1) to get B.E./B.Tech with Multidisciplinary Minor (Following structure)

Sr. No.	Course Category	Course Code	Course Title	Contact Hours per week		Credits		Scheme of Examination		
				Theory	Practical	Theory	Practical	CIA	SEE	Total
1	Program Core Course (PCC)	MED 201	Engineering Thermodynamics	3	----	3	----	40	60	100
		MED 202	Engineering Thermodynamics Lab		2		1	20	30	50
		MED 203	Machine Drawing	2	----	2	----	20	30	50
		MED 204	Machine Drawing Lab	---	2	--	1	20	30	50
		MED 205	Manufacturing Processes	2	----	2	----	20	30	50
		MED 206	Workshop Practice	---	2	--	1	20	30	50
2	Multidisciplinary Minor (MDM)	MDMT-1	Student will have to choose any one course from the Basket of minor	2	----	2	----	20	30	50
3	Open Elective (OE) to be chosen compulsorily from the faculty other than Major Discipline	OET-1	Student will have to choose any one course from the open electives	2	----	2	----	20	30	50
		OET-2	Student will have to choose any one course from the open electives	2	----	2	----	20	30	50
4	Entrepreneurship/ Economics/ Management Courses	HSM- I HSM- II HSM- III	Theory-7 One Theory course form the following Basket 1) Engineering Economics 2) Entrepreneurship Development 3) Industrial Management	2	----	2	----	20	30	50
5	Value Education Course (VEC)	VECT-1	Theory-8 (Universal Human Values/ Environmental Studies)	2	---	2	--	20	30	50
6	Comm. Engg. Project/ Field Project (FP)	FP-1	Field Project	---	4	---	2	20	30	50
				17	10	17	5	260	390	650

Option 2 : Student can choose option 2 to get B.E./B.Tech Honors with Multidisciplinary Minor (above structure + following additional Honors degree course)

7	One additional Course from core program for Honors Degree	MED 207	Additive Manufacturing	3	----	3	----	40	60	100
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Please note that the course listed under Sr. No. 7 is optional and specifically designed for students pursuing a Bachelor's Degree Honors with Multidisciplinary Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option

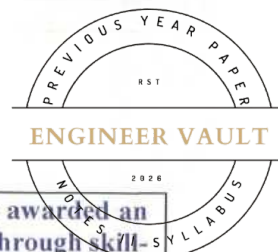
Option 3 : Student can choose option 3 to get B.E./B.Tech with Double Minor (Option 1 structure + following additional Double Minor course)

8	One additional Minor Course (for Double Minor)	DM-201	Basic Mechanical Sciences	3	----	3	----	40	60	100
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Please note that the courses listed under Sr. No. 8 is optional and specifically designed for students pursuing a Bachelor's Degree with Double Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option.

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Option 2 : Student can choose option 2 to get B.E./B.Tech Honors with Multidisciplinary Minor (above structure + following additional Honors degree course										
8	One additional Course from core program for Honors Degree	MED 257	Supply Chain Management	3	---	3	---	40	60	100
Please note that the course listed under Sr. No. 8 is optional and specifically designed for students pursuing a Bachelor's Degree Honors with Multidisciplinary Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option										
Option 3 : Student can choose option 3 to get B.E./B.Tech with Double Minor (Option 1 structure + following additional Double Minor course)										
9	One additional Minor Course (for Double Minor)	DM-251	Thermal Engineering	3	---	3	---	40	60	100
Please note that the courses listed under Sr. No. 9 is optional and specifically designed for students pursuing a Bachelor's Degree with Double Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option										



Exit Option: Students may choose to exit the program at the end of the second year and be awarded an UG Diploma in the Mechanical Engineering, provided they have earned additional 8 credits through skill based vocational courses, internships, or mini project undertaken during the summer vacation, after second year.

List of Multidisciplinary Minors from Mechanical Engineering which will be available for the students from other discipline

Semester	Course Code	Title of the Course
III	MDMT-1	Industrial Automation
IV	MDMT-2	Robotics Kinematics and Dynamics
V	MDMT-3	Sensors and Actuators for Robotics
VI	MDMT-4	Artificial Intelligence and Machine Learning in Robotics
VII	MDMT-5	PLC and SCADA
VIII	MDMT-6	Mini Project

**Dr. Babasaheb Ambedkar Marathwada University,
Chhatrapati Sambhajnagar.**

(Faculty of Science & Technology)

Syllabus of S.Y. B. E. / B. Tech.

(Mechanical Engineering)

Semester-III

Course Code: MED 201 Course: Engineering Thermodynamics Total Credits: 3 Contact Hours Theory: 03 Hrs./week, Total Contact Hours 45 hrs Maximum Marks: 100		
Prerequisite : 1. Basic understanding of physics, especially concepts like energy, work, heat, and pressure. 2. Knowledge of mathematics, including algebra, calculus, and unit conversions, is essential for solving thermodynamic equations. 3. A foundation in chemistry helps in understanding fuels and combustion. 4. Additionally, familiarity with basic engineering systems and strong analytical skills are important for applying theoretical concepts to real-world devices.		
Learning Objectives of the Course: 1. Learn about thermodynamic system and its equilibrium, understand various forms of energy - work and heat transfer. 2. Study the basic laws of thermodynamics including, zeroth law, first law and second law. 3. To understand the applications of the first and second laws of Thermodynamics to various gas processes and cycles. 4. To study the Carnot Cycle and the concept of Entropy 5. To understand the various Air standard and Vapor power cycles and their Performance.		
Course Outcomes (COs) : After completion of the course, students will be able to - 1. Understand the basic concepts of thermodynamics like systems, equilibrium, process etc. and its applications. 2. Identify the different types of work and heat transfer mechanisms. 3. Apply the laws of thermodynamics to real system. 4. Differentiate reversible and irreversible process using second law and entropy concepts. 5. Apply the theoretical knowledge of internal combustion engines to determine the performance characteristics and draw heat balance sheet of petrol and diesel engines. 6. Evaluate the performance of air standard cycles, vapor power cycles and Refrigeration cycles		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	FIRST LAW OR THERMODYNAMICS APPLIED TO FLOW PROCESS Concept of Flow work, control volume and steady flow process, assumptions, Steady flow energy equation on time and mass basis, difference between steady flow and non- flow process, study and applications of SFEE to some steady flow devices viz nozzles, diffusers,	15 Hrs

	throttling valve, turbine, compressors, I.C. Engine, Heat Exchangers etc. Limitations of First law of Thermodynamics, Concept of PMM-I (Descriptive and Numerical Treatment) SECOND LAW OF THERMODYNAMICS Various statements, Heat engine, Refrigerator and Heat pump. COP of Heat pump and Refrigerator reversed heat engine, Equivalence of Kelvin-Planck and Clausius statements, PMM-II, Carnot theorem, Thermodynamic temperature scale. (Descriptive and Numerical Treatment).	
II	ENTROPY Entropy: Introduction, Clausius theorem, T-s plot, Clausius inequality, Entropy, and Irreversibility, Entropy principle and its application, Entropy and direction, Entropy, and disorder. Availability: Available energy pertaining a cycle, Quality of energy, law of degradation of energy, maximum work in a reversible process, Dead state, Availability in steady flow and non-flow processes, second law efficiency. (Descriptive Treatment) POWER CYCLE Concept of air standard cycle, assumptions, Carnot, Otto, Diesel, and dual air standard cycles with representation on P-V & T-S planes, mathematical analysis for efficiency, mean effective pressure. and power output, comparison. Brayton cycles, Atkinson cycle, Ericsson Cycle. (Descriptive and Numerical Treatment).	15 Hrs
III	PROPERTIES OF STEAM OR PURE SUBSTANCE Pure substance, phase, phase transformation of water at constant pressure, p-v, T-s and h-s diagrams, critical point, Triple point, Different stages, Entropy of steam, steam tables, representation of processes of steam on p-v, T-s, and h-s diagrams, dryness fraction and its measurement. (Descriptive and Numerical Treatment). FUELS AND COMBUSTION: Definition of a fuel, types of fuels calorific values and its determination, combustion equation for hydrocarbon fuel on mass and volume basis, determination of minimum air required for combustion and excess air supplied, conversion of volumetric analysis to mass analysis and vice versa, fuel gas analysis, analysis of product of combustion, determination of actual and excess air quantity from combustion analysis and stoichiometric and actual air to fuel ratios. Orsat apparatus and flue gas analysis. (Descriptive and Numerical Treatment)	15 Hrs
Reference Books: 1. P. K. Nag, "Engineering Thermodynamics" 6th Edition, Tata McGraw Hill Publication 2017. 2. R.K Rajput, "Engineering Thermodynamics" Lakshmi Publications 3. Moran J Shapiro, "Fundamentals of Engineering Thermodynamics", 8th Edition, John Wiley. 4. Yunus A. Cengel and Michael A. Boles, "Thermodynamics, An Engineering Approach", 9th Edition, Tata McGraw Hill publications 5. Claus Borgnakke, Richard Edwin Sonntag, "Fundamentals of Thermodynamics", Edition, 2020, Wiley India edition		

Course Code: MED 202 Course: Engineering Thermodynamics Lab Total Credits: 1 Contact Hours Practical : 02 Hrs./week, Total Contact Hours 30 hrs Maximum Marks: 50	
Prerequisite: • Basic understanding of physics, especially concepts like energy, work, heat, and pressure Learning Objectives of the Course: 1. Familiarize students with thermodynamic systems, properties, and various thermodynamic cycles. 2. Develop hands-on skills in using instruments like calorimeters and Orsat apparatus. 3. Strengthen conceptual understanding of key thermodynamic laws and processes through experiments and assignments. 4. Bridge theory and practice via industrial exposure to real-world thermal systems. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Identify and differentiate various thermodynamic systems and properties. 2. Understand the working of basic heat engines and thermodynamic cycles. 3. Determine the calorific value of fuels using bomb and gas calorimeters. 4. Analyze the quality of steam by determining its dryness fraction. 5. Perform flue gas analysis using the Orsat apparatus to evaluate combustion efficiency.	
Topics / actual contents of the syllabus	Contact Hours
List of the Experiments 1. Study of Thermodynamic Systems and Properties 2. Study of Heat Engines and Thermodynamic Cycles 3. Study of determination of Calorific Value of Fuels by using Bomb calorimeters 4. Study of determination of Calorific Value of Fuels by using Boys Gas calorimeter calorimeters 5. Study of determination of dryness fraction of steam. 6. Study of flue gas analysis by using Orsat apparatus. 7. Assignment on First law of thermodynamics 8. Assignment on Second law of thermodynamics 9. Assignment on Entropy 10. Assignment on Availability and 11. Assignment on Power cycle 12. Assignment on Pure substances 13. Assignment on Combustion of fuel 14. Industrial visit to thermal power plant 15. Industrial visit to Ice factory	30 Hrs
Students should undertake at least 10 experiments during the semester from above list	

Course Code: MED 203 Course: Machine Drawing Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Prerequisite : 5. Knowledge of orthographic projection, Projection of Solids and drafting skills of Engineering drawing. Learning Objectives of the Course: 1. The subject intends to make the students understand various curves used in machine components and their development. 2. Interpret the industrial drawings and understand various conventions of machine components. 3. Visualize and construct the assembly of given set of individual components. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Represent any given three-dimensional object into two dimensional views by using first angle projection method. 2. Demonstrate the knowledge of drafting skills by using drawing instruments. 3. Using standard conventions sketch various machine components like fasteners, gears, and pulleys etc. along with their surface contour like threads, tolerances, levels of surface finish. 4. Classify various machine parts and their joints using standard conventions. 5. Interpret the true shape of cut section of any geometric solids like prisms, pyramids, cone, cylinder, and any other standard machine component. 6. Develop an assembly drawing using parts drawing of machine components		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Development of Surfaces Draw the development of surfaces for sections of Prisms, Cylinders, Pyramids and Cone Interpenetration of solids Draw the curves of interpenetration of the surfaces of the solids such as Cylinder, Prism, Pyramid, Cone, and Sphere.	10 Hrs
II	Auxiliary views Study of auxiliary planes, projection of objects on auxiliary planes, completing the regular views with the help of given auxiliary views. Engineering curves Draw the various curves like ellipse, Parabola, Hyperbola, Involute, Cycloid, Epicycloid, Hypocycloid and Helix.	10 Hrs

III	Conventional representations: Representation of elements of machine drawing: Engineering Materials, Surface finishes, tolerances, Different types of Screw threads. Component Drawings: Bolts and Nuts, Locking devices, Keys and Cotter joints, welded joints, Knuckle Joint, Riveted joints, Shaft Couplings, Bearings and Pipe joints.	10 Hrs
Reference Books: 1. N. D. Bhatt, "Machine Drawing", 50th Edition, Charotar Publishing House Pvt. Ltd. 2. Dr. R. K. Dhawan, "Machine Drawing", 15th Edition, S. Chand and Company Pvt. Ltd 3. N. D. Bhatt, "Engineering Drawing", 53rd Edition, Charotar Publishing House Pvt. Ltd. 4. P.S. Gill, "A Textbook of Machine Drawing", S. K. Kataria & Sons' 2013. 5. N. Sidheswar, P. Kanata and V.V.S. Sastry, "Machine Drawing", McGraw Hill Education, 2017.		

Course Code: MED 204 Course: Machine Drawing Lab Total Credits: 1 Contact Hours Practical: 02 Hrs./week, Total Contact Hours 30 hrs Maximum Marks: 50	
Prerequisite: Knowledge of orthographic projection, Projection of Solids and drafting skills of Engineering drawing. Learning Objectives of the Course: 1. The subject intends to make the students understand various curves used in machine components and their development. 2. Interpret the industrial drawings and understand various conventions of machine components. 3. Visualize and construct the assembly of given set of individual components. Course Outcomes (COs) : 1. Understand and apply standard drawing practices and conventions in mechanical drafting. (Related to: Drawing standards in the sketchbook) 2. Construct and interpret the locus of points for simple mechanisms, demonstrating knowledge of kinematic motion. (Related to: Locus of simple mechanisms and applications) 3. Develop the lateral surfaces of various solid objects and truncated shapes accurately using development methods. (Related to: Development of surfaces) 4. Analyze and represent the intersection of solids in orthographic views with precision. (Related to: Intersection of solids) 5. Create detailed component drawings from assembly drawings, identifying individual part specifications and geometric features. (Related to: Assembly to details drawing) 6. Construct complete assembly drawings from multiple part drawings, demonstrating understanding of fits, tolerances, and assembly sequence. (Related to: Details to assembly drawing)	
Topics / actual contents of the syllabus	Contact Hours
List of the Experiments 1. Drawing Standards in sketchbook. 2. Using full size sheet (A-1) draw two problems of each on unit I, II and III. 3. Using full size sheet (A-1) draw two problems on different types of curves. 4. Using full size sheet (A-1) draw conventions of various machine components. 5. Using full size sheet (A-1) draw Assembly drawing of at least one assembly mentioned in syllabus. 6. Using full size sheet (A-1) draw details drawing of at least one assembly mentioned in syllabus.	30 Hrs

Course Code: MED 205 Course: Manufacturing Processes Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours 30 hrs Maximum Marks: 50		
Prerequisite : 1. Basic knowledge about mechanical engineering 2. Fundamental knowledge of mechanical properties and types of forces Learning Objectives of the Course: 1. To develop an understanding of the principles, classification, and applications of various manufacturing processes. 2. To provide insights into non-conventional machining processes, their need, benefits, mechanisms, and industrial applications. 3. To equip students with practical knowledge of different manufacturing techniques, ensuring an ability to select appropriate processes for different engineering applications. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Recall classification, advantages, and disadvantages of various manufacturing processes. 2. Explain the working principle of different manufacturing processes. 3. Describe construction, working and specifications of machine tools required for manufacturing. 4. Identify need of various non-conventional manufacturing processes 5. Choose the appropriate manufacturing processes for producing a given component. 6. Identify various causes of defects in manufacturing processes and suggest suitable remedies		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Casting and Forming processes. Introduction and classification of metal casting processes, Sand casting processes: pattern, core, mold, and mold making, Expendable mold casting processes, Permanent mould casting processes, Casting defects, Types of Furnaces: Cupola, Crucible, Electric arc and Induction. Introduction and classification of metal forming operations, material behavior in metal forming, rolling processes, Forging processes, Extrusion processes, Wire drawing, Bar drawing, Tube drawing.	10 Hrs

II	Machining processes Turning: Operations related to turning, Cutting tools, Types of lathes, Calculations for machining time. Boring: Boring operation and machines, Calculations for machining time. Drilling: Operations related to drilling, Geometry of twist drill, Drill machine operation and types, Calculations for machining time in drilling operation. Milling: Operations related to milling, cutting tools and cutting conditions in milling, Milling machine operation and types, Calculations for machining time in milling. Grinding: Grinding operation types, Grinding machine operations and types, Grinding wheel material, grain size, bonding materials, wheel structure and wheel grade, Grinding wheel specification	10 Hrs
III	Metal joining and non-conventional process Welding: classification of welding processes, types of joints and welds, HAZ, Arc welding, Resistance welding, Oxy-fuel gas welding, Solid state-welding, welding defects. Brazing and soldering processes Non-conventional machining processes: Need, benefits, classification, mechanism of metal removal, advantages, disadvantages, and applications of EDM, ECM, LBM, USM, AJM, WJM.	10 Hrs
Reference Books: 1. Hajra Chaudhary, "Workshop Technology", Vol. 1 & 2, 10th Edition, Dhanpat Rai and Sons. 2. Amitabh Ghosh, "Manufacturing Science", Vol. 1 & 2, 2nd Edition, East-West Press. 3. R. K. Jain, "Production Technology", Vol. 1 & 2, 17th Edition, Khanna Publications. 4. Mikell P. Groover, "Fundamentals of Modern Manufacturing", Vol. 1, 4th Edition, John Wiley & Sons.		

Course Code: MED 206 Course: Workshop Practice Total Credits: 1 Contact Hours Practical: 02 Hrs./week, Total Contact Hours 30 hrs Maximum Marks: 50	
Learning Objectives of the Course: 1. To have hands on practice and understanding of moulding process and tools. 2. To have hands on practice and understanding of welding process and tools. 3. To have hands on practice and understanding of turning process and tools. 4. To have hands on practice and understanding of machining process and tools. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Identify workshop tools and their operational capabilities. 2. Practice manufacturing components using workshop trades including molding, welding, turning, drilling and milling operations 3. Apply suitable tools for machining processes including turning, facing, thread cutting and tapping. 4. Demonstrate the use of measurement tools to manufacture the job.	
Topics / actual contents of the syllabus	Contact Hours
List of the Experiments 1. Foundry Study of sand moulding, Types of sand and moulding equipment. List of various operations and tools. Workshop diary – Draw sketches and description of different moulds and of job. Practical: One job of moulding (Single or multi -piece pattern). 2. Welding Study of arc welding machines, MIG welding machine, TIG welding machine and welding equipment. List of various operations and tools used. Workshop diary – Sketch of job. Practical: One job of welding individually of any useful item of daily use using any welding operation. 3. Turning Shop Study of Different operations to be carried on lathe machine using tail stock, taper turning methods (Calculations), internal and external threading, facing finishing cuts. Workshop diary – Sketch of job, List of various operations and tools Practical: Preparing one jobs on lathe machine performing all above operations. 4. Machine Shop Study of different types of drilling and milling machines and different types of operations carried out on them. List of various operations and tools used. Workshop diary – Sketch of job involving drilling and milling operations.	30 Hrs

Practical: Preparing a job individually or in a group of students i.e. Gear blank by turning, external milling of gear teeth involving calculations for indexing.	
Students should undertake at least 3 practical during the semester from above list	

Course Code: MDMT-I Course: Industrial Automation (This course will be available for the students from other discipline and Mechanical Engineering students will choose minor course from other discipline) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Prerequisite : 1. Basic knowledge of electrical and electronics engineering. 2. Fundamentals of control systems and instrumentation. Learning Objectives of the Course: 1. To introduce the importance of automation techniques manufacturing and process industries. 2. To impart the role of PLC in industry automation. 3. To expose to various control techniques employed in process automation. 4. To develop automation system for manufacturing and process industries Course Outcomes (COs) : After completion of the course, students will be able to - 1. Define key concepts of industrial automation, including automation principles, system components, and control strategies. 2. Explain the role of automation in modern industries, including Industry 4.0 and smart manufacturing. 3. Utilize sensors, actuators, and power electronic devices in industrial control systems. 4. Examine the different industrial communication protocols such as MODBUS, PROFIBUS, and Ethernet/IP for efficient automation. 5. Assess the performance and suitability of PLCs, SCADA, and DCS for different industrial applications. 6. Design and implement basic PLC-based automation systems integrated with SCADA and HMI for real-world applications.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Industrial Automation Introduction to Automation: Principles and Strategies of Automation, Basic Elements of an Automated System, Advanced Automation Functions, Levels of Automations, Types of Automation: Fixed, Programmable, and Flexible, Role of Automation in Industry 4.0. Material handling and its function Types of Material Handling Equipment Automated Flow lines, Methods of Work-part Transport, Transfer Mechanism, Buffer Storage, Control Functions, and Automation for Machining Operations, Design and Fabrication Considerations.	10 Hrs
II	Components of Industrial Automation Sensors for temperature, pressure, force, displacement, speed, flow, level, humidity, and pH measurement. Actuators, process control valves, power electronics devices DIAC, TRIAC, power MOSFET and IGBT. Introduction of DC and AC servo drives for motion control System. Programmable Logic Controllers (PLCs)	10 Hrs

	Programmable Logic Controllers (PLCs) - Introduction to PLC, Logic Functions, Input & Output Modules, PLC Processors, PLC Instructions, Introduction to Programming a PLC.	
III	Industrial Communication SCADA, HMI, and Control System Industrial Communication Protocols: MODBUS, PROFIBUS, Ethernet/IP, Supervisory Control and Data Acquisition (SCADA), Human-Machine Interface (HMI) Systems, Integration of PLCs with SCADA & HMI, Case Studies on Industrial Automation Applications Distributed Control System: Overview of DCS, DCS software configuration, DCS communication, DCS Supervisory Computer Tasks, DCS integration with PLC and Computers, Features of DCS, Advantages of DCS.	10 Hrs
Reference Books: 1. A.K. Gupta, S.K. Arora, "Industrial Automation and Robotics", Vol. 1, Latest Edition, Laxmi Publications, 2022. 2. Mikell P. Groover, "Automation, Production Systems, and Computer-Integrated Manufacturing", Vol. 1, 4th Edition, Pearson, 2015. 3. Stamatio Manesis, George Nikolakopoulos, "Introduction to Industrial Automation", Vol. 1, Latest Edition, CRC Press, 2018. 4. Godfrey C. Onwubolu, "Mechatronics: Principles and Applications", Vol. 1, Latest Edition, Elsevier, 2017.		

Entrepreneurship/Economics/Management Courses

Students will have to choose any one theory course form the following Basket. Any
One course in 3rd semester and another course in 4th semester

- 1) Engineering Economics (EEM-I)
- 2) Entrepreneurship Development (EEM-II)
- 3) Industrial Management (EEM-III)

Course Code: EEM-I Course: Engineering Economics (Semester III/ IV)		
Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Learning Objectives of the Course: 1. Understand the Time Value of Money 2. Evaluate Engineering Alternatives 3. Analyze Public Sector Projects 4. Understand and Apply Depreciation and Taxation 5. Perform Break-even and Sensitivity Analysis 6. Understand Cost Concepts for Decision-Making		
Course Outcomes (COs) : After completion of the course, students will be able to - 1. To evaluate Time Value of Money 2. To enable students to perform economic comparisons between different projects or assets 3. To introduce benefit-cost analysis for evaluating the economic feasibility of projects 4. To apply break-even analysis using both linear and non-linear models 5. To develop skills in cost estimation, cost control, cost reduction, and identify relevant costs for decision-making.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Time value of money: Simple and compound interest, Time value equivalence, Compound interest factors, Cash flow diagrams, Calculation, Calculation of time – value equivalences. Present worth comparisons, Comparisons of assets with equal, unequal and infite lives, comparison of deferred investments, Future worth comparison, payback period comparison.	10 Hrs
II	Use and situations for equivalent annual worth comparison, Comparison of assets of equal and unequal life. Rate of return, Internal rate of return, comparison of IIR with other methods, IRR misconceptions. Analysis of	10 Hrs

	public Projects: Benefit/ Cost analysis, quantification of project, cost and benefits, benefit/ cost applications, Cost effectiveness analysis.	
III	Depreciation , Computing depreciation charges, after tax economic comparison, Break-even analysis; linear and non-linear models. Sensitivity analysis: single and multiple parameter sensitivity. Fixed and variable cost , Product and Process Costing, Standard Costing, Cost estimation, Relevant Cost for decision making, Cost estimation, Cost control and Cost reduction Techniques.	10 Hrs

Reference Books:

1. Riggs, J.L., Dedworth, Bedworth, D.B., Randhawa, S.U., "Engineering Economics", Vol. 1, Latest Edition, McGraw Hill International, 1996.
2. James L. Riggs, David D. Bedworth, Sabah U. Randhawa, "Economics for Engineers", Vol. 1, 4th Edition, McGraw-Hill, 2004.
3. Donald Newnan, Ted Eschembach, Jerome Lavelle, "Engineering Economic Analysis", Vol. 1, 8th Edition, Oxford University Press, 2012.
4. John A. White, Kenneth E. Case, David B. Pratt, "Principles of Engineering Economic Analysis", Vol. 1, 6th Edition, John Wiley, 2010.
5. R. Panerseeelvm, "Engineering Economics", Vol. 1, 2nd Edition, PHI, 2008.
6. Michael R. Lindeburg, "Engineering Economics Analysis", Vol. 1, Latest Edition, Professional Publications, 1993.
7. V. Mote, S. Paul, G. Gupta, "Managerial Economics", Vol. 1, Latest Edition, Tata McGraw Hill, 2004.

Course Code: EEM-II Course: Entrepreneurship Development (Semester III/ IV) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Learning Objectives of the Course: 1. Understand the fundamentals of entrepreneurship and its significance in engineering. 2. Identify business opportunities and develop innovative ideas. 3. Create a basic business plan and understand key entrepreneurial strategies. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Explain the concept of entrepreneurship and its importance in the modern economy. 2. Identify and evaluate business opportunities in engineering and technology sectors. 3. Develop a comprehensive business plan including financial, operational, and marketing strategies. 4. Assess the challenges and risks in entrepreneurship and develop strategies to mitigate them. 5. Demonstrate entrepreneurial thinking through case studies, projects, and presentations.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Entrepreneurship Definition, meaning, and characteristics of an entrepreneur, Evolution of entrepreneurship Types of entrepreneurs, Role of entrepreneurship in economic development, Key traits of successful entrepreneurs.	10 Hrs
II	Opportunity Recognition and Idea Generation Identifying business opportunities, Creativity and innovation in entrepreneurship, Techniques for idea generation, Feasibility analysis (technical, market, financial).	10 Hrs
III	Business Planning Components of a business plan, Business models and strategy formulation, Legal requirements for starting a business, Intellectual Property Rights (IPR) and patents.	10 Hrs
Reference Books: 1. C.B. Gupta, Srinivasan, "Entrepreneurship Development", Vol. 1, Latest Edition, Sultan Chand & Sons, 2020. 2. Donald F. Kuratko, "Entrepreneurship: Theory, Process, Practice", Vol. 1, Latest Edition, Cengage, Eric Ries, "The Lean Startup", Vol. 1, Latest Edition, Anonymous, 2011. 3. Alexander Osterwalder, Yves Pigneur, "Business Model Generation", Vol. 1, Latest Edition, Wiley, 2011. 4. Peter F. Drucker, "Innovation and Entrepreneurship", Vol. 1, Latest Edition, Taylor & Francis, 2014.		

Course Code: EEM-III Course: Industrial management (Semester III/ IV) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Learning Objectives of the Course: 1. To understand concept of management, administration, Organization, Industrials Laws. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Understand the fundamental principles of management. 2. Describe different forms of business organizations and organizational structures. 3. Apply theoretical knowledge to real-world management and organizational challenges. 4. Analyze the impact of organizational, environmental factors. 5. Explain the role of economics in management decision-making.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Management: Managing and Manager, organizations and need for management, the managing process, types of managers, the challenge of management, the evolution of Management theory. Management in the 21st century: The importance of organizational and natural environment, elements of direct action environment, managing multiple stock holder relationship, elements of the indirect action environment, Natural Environment management 2000 and beyond, social responsibility and ethics, globalization. Evaluation of case studies related to above concept.	10 Hrs
II	Business Organization Forms of business organization, individual proprietorship, joint stock company, co-operative enterprise, co-operative enterprise and public sector undertakings. Organization structure in industries, Line organization, functional organization, line and staff organization, committee organization, project organization matrix organization. Nature and Significance of Economics. Science, engineering and technology, their relationship with economic development. Basic economic concepts, human wants economic goods, utility value, price cost, wealth and capital. Demand supply, elasticity of demand and supply. Concept of profit and revenues.	10 Hrs
III	Accidents and safety Classification of accidents, according to nature of industries; i.e. fatal, temporary, according to event and place. Causes of accidents , psychological, and other industrial hazards. Effects of accidents.	10 Hrs

	Accident-prone workers, accident to be taken incase of accidents with machines, electric shock, road accident fires and erection and construction accidents Personnel Management: Man power, sources of recruitment, selection and training, job evaluation, performance appraisal, wages and incentives, self and time management.	
Reference Books: 1. James A. F., "Management", Vol. 1, 6th Edition, PHI. 2. Claude S. George, Jr., "Management for Business and Industry", Vol. 1, Revised Edition, Prentice-Hall of India Private Limited. 3. McConnell, Gupta, "Economics: Principles, Problems, and Policies", Vol. 1, 18th Edition, The McGraw-Hill. 4. T.R. Banga, S.C. Sharma, "Industrial Organisation and Engineering Economics", Vol. 1, Latest Edition, Khanna Publishers, Jan-2006. 5. O.P. Khanna, "Industrial Engineering & Management", Vol. 1, Latest Edition, Dhanpat Rai Publication, Jan-2018.		

Value Education Courses (VEC)

Students will have to choose any one theory course form the following Basket. Any
One course in 3rd semester and another course in 4th semester

- 1) Universal Human Values (VECT-I)
- 2) Environmental Students (VECT-II)

Course Code: VECT-I Course: Universal Human Values (Semester III/ IV) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50 Learning Objectives of the Course: - To appreciate the essential complementarities between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity, which are the core aspirations of all human beings, - To facilitate the development to a holistic perspective among students to lead their Personal and professional lives in an ethical way. - To highlight plausible implications of such a holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour, and mutually enriching interaction with nature. Course Outcomes (COs) : After completion of the course, students will be able to - - Define key terms related to human values. - Explain the concept of happiness as related to right understanding and relationship. - Apply the principles of right understanding in their daily interactions. - Analyze the impact of their values on their behaviour and decisions.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Value Education and Harmony in the Human Being - Understanding Value Education - Self-exploration as the Process for Value Education - Continuous Happiness and Prosperity - the Basic Human Aspirations and their fulfillment - Right Understanding, Relationship and Physical Facility - Happiness and Prosperity - Current Scenario - Method to Fulfill the Basic Human Aspirations - Harmony in the Human Being - Understanding Human being as the Co-existence of the Self and the Body - Distinguishing between the Needs of the Self and the Body - the Body as an instrument of the Self	10 Hrs

	• Understanding Harmony in the Self • Harmony of the Self with the Body • Programme to ensure self-regulation and Health.	
II	Harmony in the Family and Society • Harmony in the Family - the Basic Unit of Human Interaction "Trust" - the Foundational Value in Relationship • 'Respect' - as the Right Evaluation Values in Human-to-Human Relationship • Harmony in the Society • Other Feelings, Justice in Human-to-Human Relationship • Understanding Harmony in the Society • Vision for the Universal Human Order	10 Hrs
III	Harmony in the Nature (Existence) and Implications of the Holistic Understanding • Understanding Harmony in the Nature • Interconnectedness, self-regulation, and Mutual Fulfillment among the Four Order of Nature • Realizing Existence as Co-existence at All Levels • The Holistic Perception of Harmony. Implications of the Holistic Understanding - a Look at Professional Ethics • Basis for Universal Human Values • Definitiveness of (Ethical) Human Conduct • Professional Ethics in the light of Right Understanding • A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order • Holistic Technologies, Production Systems and Management Models, Typical Case Studies Strategies for Transition towards Value-based Life and Profession.	10 Hrs

Reference Books:

1. P.L. Dhar, R.R. Gaur, "Science and Humanism", Vol. 1, 1st Edition, Commonwealth Publishers.
2. Nagaraj, "Jeevan Vidya: Ek Pariechaya", Vol. 1, Latest Edition, Jeevan Vidya Prakashan, Amarkantak, 1999.
3. A.N. Tripathy, "Human Values", Vol. 1, Latest Edition, New Age International Publishers, 2003.
4. E.G. Seebauer, Robert L. Berry, "Fundamentals of Ethics for Scientists & Engineers", Vol. 1, 1st Edition, Oxford University Press.
5. M. Govindrajan, S. Natrajan, V.S. Senthil Kumar, "Engineering Ethics and Human Values", Vol. 1, 1st Edition, Prentice Hall of India Ltd.
6. B.P. Banerjee, "Foundations of Ethics and Management", Vol. 1, Latest Edition, Excel Books, 2005.
7. B.L. Bajpai, "Indian Ethos and Modern Management", Vol. 1, Reprinted Edition, New Royal Book Co., Lucknow, 2008.

E-resources:

1. <http://madhyasth-darshan.info/postulations/knowledge/knowledge-of-humane-conduct/>
2. https://www.youtube.com/channel/UCQxWr5QB_eZUnwxSwxXEkQw
3. <https://youtu.be/OgdNx0X923I>
4. <https://fdp-si.aicte-india.org/UHV-II%20Practice%20Sessions.php>
5. <https://fdp-si.aicte-india.org/download.php#1/>

Course Code: VECT-I Course: Environmental Studies (Semester III/ IV) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Prerequisites: Understanding of the Concept of Environment Learning Objectives of the Course: 1. To study the environment and ecosystems. 2. To study different types of natural resources. 3. Knowledge and concept of biodiversity and its conservation. 4. Basic knowledge and concept of causes, effects, and control of different types of Environmental pollution. 5. To study population growth and its impact on the environment Course Outcomes (COs) : After completion of the course, students will be able to - 1. Define key environmental terms and concepts (e.g., ecosystem, biodiversity, pollution). 2. Explain the causes and effects of major environmental problems. 3. Apply environmental principles to analyze real-world scenarios. 4. Differentiate between various types of pollution and their impacts.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to environmental studies and natural resources: Definition, scope and Importance and need for public awareness. Natural resources: Forest resources: Use and over-exploitation, deforestation. Timber extraction. Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dam's benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using Mineral resources Food, energy, and land resources: Food resources: world food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems water logging, salinity. Energy resources: Growing energy needs, renewable and non-renewable energy sources, and use of alternate energy sources. Land resources: Land as a resource, land degradation, man-induced landslides, soil erosion, and desertification.	10 Hrs
II	Ecosystems and Biodiversity and its conservation Ecosystems: Concept of an ecosystem. Structure and function of an ecosystem. Producers, consumers, and decomposers. Energy flow in the ecosystem. Ecological succession. Food chains, food webs, and ecological pyramids. Introduction, types, characteristic features, structure, and function of the following ecosystems: Forest ecosystem,	10 Hrs

	Grassland ecosystem, Desert ecosystem, and Aquatic ecosystems (ponds, streams, lakes, rivers, Oceans, estuaries) Biodiversity and its conservation: Introduction Definition: genetic, species, and Ecosystem diversity. Bio geographical classification of India. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity	
III	Environmental Pollution and Social issues and the Environment Environmental Pollution: Definition, Cause, effects, and control measures of Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, nuclear hazards, Role of an individual in the prevention of pollution. Social issues and the Environment: From Unsustainable to sustainable development Urban problems related to energy. Climate change, global warming, acid rain, ozone layer depletion Environment Protection Act. Public awareness.	10 Hrs
Reference Books: 1. Agarwal K.C., "Environmental Biology", Vol. 1, Latest Edition, Nidi Puhl Ltd., Bikaner, 2001. 2. Bharucha Erach, "The Biodiversity of India", Vol. 1, First Edition, Mapin Publishing Pvt. Ltd., Ahmedabad. 3. Heywood V.H., Waston, "Global Biodiversity Assessment", Vol. 1, Latest Edition, Cambridge University Press, 1995. 4. Jadhav H., Bhosale V.M., "Environmental Protection and Laws", Vol. 1, First Edition, Himalaya Publishing House, Delhi. 5. Odum E.P., "Fundamentals of Ecology", Vol. 1, First Edition, W.B. Saunders Co., USA. 6. Miller T.G. Jr., "Environmental Science", Vol. 1, First Edition, Wadsworth Publishing Co.		

Course Code: FP-I Course: Field Project Total Credits: 2 Contact Hours Practical: 04 Hrs./week, Total Contact Hours: 60 hrs Maximum Marks: 50		
Prerequisites: Basic knowledge of mechanical engineering principles, exposure to technical writing and research methods Learning Objectives of the Course: 1. To enable students to identify a relevant mechanical engineering problem. 2. To conduct a comprehensive literature survey using authentic research sources. 3. To formulate a well-defined problem statement. 4. To understand and define project objectives, scope, methodology, and expected outcomes. 5. To develop a strategic plan and timeline for execution in field project. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Identify a relevant and feasible mechanical engineering problem through field interaction, experimentation, or simulation. 2. Conduct a comprehensive literature review to establish the state of the art in the chosen domain. 3. Formulate a clear and specific problem statement with well-defined objectives and scope. 4. Choose appropriate tools and methodologies suited to the project requirements. 5. Develop a comprehensive project execution plan including resource allocation and risk mitigation. 6. Prepare and present effective interim and final project reports, demonstrating clarity of thought and technical writing skills.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Engineering Research Nature and scope of research in mechanical engineering Types of projects (Experimental, Computational, Design-based, Review, Industrial) Case studies of past projects Identifying the Problem Statement How to select a relevant and feasible problem Techniques for problem identification (e.g., field visits, industry interaction, lab experimentation, simulation need) Problem relevance and innovation Literature Survey Understanding scholarly sources (journals, conference proceedings, patents) How to read, analyze, and summarize research papers Using databases: Google Scholar, ScienceDirect, IEEE Xplore, ASME Digital Library Tools: Reference managers (Zotero, Mendeley)	20 Hrs
II	Problem Statement Formulation Structuring a strong and specific problem statement Defining objectives and scope	20 Hrs

	Identifying research gaps Setting measurable goals and expected outcomes Strategy & Methodology Planning Overview of research and design methodologies (experimental, analytical, simulation) Tool selection and planning: CAD/CAE tools, MATLAB, ANSYS, SolidWorks, etc. Resource planning: Materials, lab access, software, Risk assessment and mitigation strategies	
III	Project Planning & Timeline Development Work breakdown structure (WBS) Gantt chart preparation Milestones and deliverables Roles and responsibilities in team projects Interim Reviews and Final Reporting Presentation of problem statement, literature review, and methodology Interim report preparation Peer and faculty feedback	20 Hrs
Deliverables: • Problem Identification Report • Literature Review Summary • Finalized Problem Statement with Objectives and Scope • Methodology and Strategy Document • Timeline (Gantt Chart or similar) • Final Project Phase-I Report and Presentation		

Honors Degree Course

Please note that this course is optional and specifically designed for students pursuing a Bachelor's Degree Honors with Multidisciplinary Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option

Course Code: MED 207 Course: Additive Manufacturing (Honors) (Semester-III)		
Total Credits: 3 Contact Hours Theory: 03 Hrs./week, Total Contact Hours 45 hrs Maximum Marks: 100		
Prerequisite : 1. Basic understanding of manufacturing processes.		
Learning Objectives of the Course: 1. To explain the principles, key terminologies, working and applications of major AM technologies. 2. To demonstrate knowledge of design considerations, slicing techniques, and post-processing methods for AM-produced components. 3. To identify and evaluate real-world applications of AM in various industries.		
Course Outcomes (COs) : After completion of the course, students will be able to - 1. Recall the fundamental concepts, terminologies, and principles of Additive Manufacturing (AM). 2. Describe various additive manufacturing processes and their working principles, including material selection and processing techniques. 3. Utilize appropriate design considerations and slicing techniques to prepare digital models for additive manufacturing. 4. Identify post-processing methods to improve the surface quality, mechanical properties, and functionality of AM components. 5. Compare different AM technologies and materials based on their characteristics, advantages, and limitations for diverse applications. 6. Identify the role of AM in various industries and its potential impact on production efficiency.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Technologies and Processes of Additive Manufacturing (AM) Foundation of Additive Manufacturing (AM), Definitions of terms used in AM, Difference between Additive and Subtractive Manufacturing. Fused Deposition Modeling (FDM), Stereo lithography & Digital Light Processing (SLA & DLP), Selective Laser Sintering (SLS), Material	15 Hrs

	Jetting (PolyJet), Direct Metal Laser Sintering & Selective Laser Melting (DMLS & SLM), Binder Jetting.	
II	Pre-processing and Post-processing in AM Pre-processing: Introduction to design in AM, Part design considering requirements for various AM devices, Supports in AM design, Slicing techniques, STL file format, Conversion of CAD file to STL, Understanding process algorithm of slicing software and slicing techniques, Understand Honeycomb structure, Understand different resolution selection. Post-processing: Different post processing techniques like sanding, water cleaning, dry cleaning, deburring, curing, painting, vapor polishing, etc.	15 Hrs
III	Materials in AM and Applications Comparison of different AM processes and material performances with respect to strength, finish, precision, application, etc. Introduction to materials like Polymers, Metals, Ceramics, Composites for AM. Different application areas like Functional prototypes, Health care products, Dental and medical implants, Architecture, Automotive, Aerospace, Consumer home products, Toys and Gadgets, Art and Education.	15 Hrs
Reference Books: 1. Ian Gibson, David W. Rosen, Brent Stucker, "Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing", 1st Edition, Springer. 2. Patri K. Venuvinod, Weiyin Ma, "Rapid Prototyping: Laser-based and Other Technologies", 1st Edition, Springer. 3. D.T. Pham, S.S. Dimov, "Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling", 1st Edition, Springer. 4. Amit Bandyopadhyay, Susmita Bose, "Additive Manufacturing", 2nd Edition, CRC Press, Taylor and Francis Group.		

Double Minor Course

(This course will be available for the students from other discipline and Mechanical Engineering students will choose double minor course from other discipline)

Please note that this course is optional and specifically designed for students pursuing a Bachelor's Degree with Double Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option.

Course Code: DM-201 Course: Basic Mechanical Sciences (Double Minor) (Semester-III)		
Total Credits: 3 Contact Hours Theory: 03 Hrs./week, Total Contact Hours 45 hrs Maximum Marks: 100		
Prerequisite : 1. Fundamental knowledge of Physics & Mathematics. 2. Basic understanding of Materials and its properties.		
Learning Objectives of the Course: 1. Understand the basic concepts and applications of Thermal, Design, Production, and Industrial Engineering, along with the principles of thermodynamics, fluid mechanics, and heat transfer. 2. Learn about various engineering materials, their types, properties, and real-world applications, as well as the fundamentals of machine design and commonly used mechanical elements. 3. Gain knowledge of different manufacturing processes, including machine tools and surface finishing methods, and get introduced to modern tools like CAD/CAM used in mechanical engineering. 4. Understand the principles of Industrial Engineering, including work study and project planning tools like CPM and PERT, and explore emerging technologies such as Robotics, Electric Vehicles, AI, Machine Learning, Industry 4.0, and IoT		
Course Outcomes (COs) : After completion of the course, students will be able to - 1. Identify and classify various engineering materials and explain their properties, along with understanding the principles of thermodynamics, fluid mechanics, and heat transfer. 2. Understand the fundamentals of machine design and the function of key mechanical components, including the basics of CAD/CAM tools. 3. Describe different manufacturing processes, the working of machine tools, and the importance of surface finishing in production engineering. 4. Apply basic concepts of industrial engineering, including work study, project planning tools like CPM and PERT, and demonstrate awareness of emerging technologies such as AI, Robotics, and IoT.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours

I	Engineering Materials Scope & applications of Engineering Materials (Introduction, Classification, Properties) Thermal Engineering Scope & applications of Thermal Engineering, laws of Thermodynamic, Thermodynamic properties, Thermodynamic processes and its types, fluid mechanics- physical properties of fluid, Definition of heat, modes of heat transfer, Laws governing the modes of heat transfer.	15 Hrs
II	Machine Design and Theory of Machines Introduction to - Design of machine elements, Computer Aided Design, Introduction to Theory of machines- Kinematics, Dynamics, Kinetics and Statics mechanisms, Power Transmission Elements - Belt, Pulleys, Gears, Bearings, Keys and Coupling, Clutches, Fasteners and Bearings. Production Engineering and Manufacturing Processes Scope & applications of Production Engineering, Manufacturing Processes – subtractive, additive Machine Tools – (All machine tools to be studied with respect to Working principle, Block diagram, Specification and Different operations performed), Surface finishing processes, Introduction to Computer Aided Manufacturing.	15 Hrs
III	Industrial Engineering Scope & applications of Industrial Engineering - work study, Introduction to Scientific management, Operations Research- CPM, PERT. Advanced Technologies in Mechanical Engineering Advance technology in Mechanical Engineering: Introduction to – Robotics, E-vehicles, Artificial Intelligence and Machine Learning, Industry 4.0-principles, Internet of things and concepts.	15 Hrs

Reference Books:

1. P.K. Nag, "Fundamentals of Classical Thermodynamics", 8th Edition, Tata McGraw Hill.
2. Y. Cengel, M. Boles, "Thermodynamics: An Engineering Approach", 5th Edition, McGraw Hill.
3. R.K. Rajput, "Thermal Engineering", 10th Edition, Laxmi Publications.
4. D.S. Kumar, "Engineering Thermodynamics: Principles and Practices", 2nd Edition, Katsons Publications.
5. Hajra Choudhary, "Workshop Technology", Media Promoters, 2008.
6. Amitabha Ghosh, A.K. Malik, "Manufacturing Science", 2nd Edition, East West Press.
7. P.N. Rao, "Manufacturing Technology", Tata McGraw Hill, 2000.
8. S.K. Garg, "Comprehensive Workshop Technology", 3rd Edition, Laxmi Publications.
9. Ce Zhang, Jianming Yang, "History of Mechanical Engineering", 1st Edition, Springer International Publishing, 2000.
10. U. Dixit, M. Hazarika, J. Davim, "A Brief History of Mechanical Engineering", 1st Edition, Springer International Publishing, 2017.
11. Praveen Kumar, "Basic Mechanical Engineering", 2nd Edition, Pearson Education, 2018.
12. Saeed Movaveni, "Engineering Fundamentals: An Introduction to Engineering", 4th Edition, Cengage Learning India Pvt. Ltd., 2011.

13. Philip Kosky, Robert Balmer, William Keat, George Wise, "Exploring Engineering: An Introduction to Engineering and Design", 4th Edition, Academic Press, 2015.
14. D.S. Hira, "Operations Research", S. Chand Publishing, 1992.
15. E. Zimmers, M. Groover, "CAD/CAM: Computer-Aided Design and Manufacturing", Pearson, 2014.

Websites and online courses:

1. https://onlinecourses.nptel.ac.in/noc21_me59/preview
2. https://onlinecourses.nptel.ac.in/noc23_me31/preview
3. <https://archive.nptel.ac.in/courses/107/101/107101087/>
4. <https://archive.nptel.ac.in/courses/112/104/112104304/>
5. <https://archive.nptel.ac.in/courses/112/107/112107292/>
6. <https://archive.nptel.ac.in/courses/112/105/112105249/>
7. <https://nptel.ac.in/courses/106105195>
8. https://onlinecourses.nptel.ac.in/noc22_cs24/preview
9. <https://nptel.ac.in/courses/106106139>

**Dr. Babasaheb Ambedkar Marathwada University,
Chhatrapati Sambhajnagar.**

(Faculty of Science & Technology)

Syllabus of S.Y. B. E. / B. Tech.

(Mechanical Engineering)

Semester-IV

Course Code: MED 251 Course: Fluid Mechanics and Fluid Machines Total Credits: 3 Contact Hours Theory: 03 Hrs./week, Total Contact Hours 45 hrs Maximum Marks: 100		
Learning Objectives of the Course: 1. Define fluid and calculate various properties of fluid. 2. Calculate hydrostatic forces and analyze stability of floating bodies. 3. Understand and calculate fluid motion and flow types. 4. Apply Bernoulli's and Navier-Stokes equations to solve fluid mechanics problems. 5. Analyze laminar and turbulent flows in engineering systems. 6. Apply dimensional analysis for modeling and experimentation. 7. Understand working principles of hydraulic turbines and centrifugal pumps. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Define fluid properties and fundamental concepts of fluid statics. 2. Analyze fluid motion using continuity, velocity potential, and stream function. 3. Apply Bernoulli's and momentum equations to practical fluid flow problems. 4. Perform dimensional analysis and apply model testing in engineering. 5. Evaluate boundary layer characteristics and flow separation. 6. Analyze the working and performance of hydraulic machines and pumps.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Basics of Fluid Mechanics and Fluid Statics Definition and properties of fluids, types of fluids: ideal vs real, Newtonian, and non-Newtonian, compressible and incompressible. Viscosity, surface tension, capillarity, pressure in fluids, Pascal's law, manometry. Hydrostatic force on immersed surfaces, center of pressure, buoyancy, stability of floating bodies, metacentric height. Engineering applications of fluid statics. Fluid Kinematics Description and types of fluid motion, velocity, and acceleration of fluid particles. Flow lines: streamlines, path lines, and streamlines. Continuity equation (Cartesian and polar), circulation and vorticity, velocity potential, stream function, relationship between them	15 Hrs

II	Fluid Dynamics Euler's and Bernoulli's equations of motion and their assumptions. Applications of Bernoulli's theorem. Momentum equation and its use in determining forces (e.g., on pipe bends). Dimensional Analysis Dimensional homogeneity, Buckingham π-theorem, formation, and interpretation of dimensionless groups. Types of similarities: geometric, kinematic, and dynamic. Flow through Pipes Loss of energy in pipes, Darcy-Weisbach equation, major and minor losses, Hydraulic Gradient Line (HGL) and Total Energy Line (TEL), flow through series pipes, parallel pipes and branched pipes, equivalent pipe	15 Hrs
III	Hydraulic Machines Impact of Jet: Forces on flat and curved plates (fixed/moving), hinged plates, and vanes. Hydraulic Turbines: Introduction, classification, working principles and design of Pelton wheel, Francis's turbine, Kaplan turbine. Draft tube and performance characteristics Centrifugal Pumps: Types, types of casing, manometric head, efficiencies, power requirements, minimum starting speed. Multi-stage pumps and performance analysis.	15 Hrs
Reference Books: 1. S.K. Agrawal, "Fluid Mechanics and Hydraulic Machines", Vol. 1, Latest Edition, IMH Publication. 2. Dr. R.K. Bansal, "Fluid Mechanics and Hydraulic Machinery", Vol. 1, Latest Edition, Laxmi Publication. 3. Dr. Jagdish Lal, "Hydraulic Machines", Vol. 1, Latest Edition. 4. Modi and Seth, "Fluid Mechanics and Hydraulic Machines", Vol. 1, Latest Edition. 5. Dr. K.K. Sudevan, "Reversible Axial Flow Pump Turbine", Vol. 1, Latest Edition. 6. Subramanya, "Fluid Mechanics and Hydraulic Machines", Vol. 1, Latest Edition, TMH Publication. 7. Shames, "Fluid Mechanics", Vol. 1, Latest Edition. 8. Streeter and Wylie, "Fluid Mechanics", Vol. 1, Latest Edition. 9. Dr. Subirkar, "Fluidic Logic & Control", Vol. 1, Latest Edition.		

Course Code: MED 252 Course: Fluid Mechanics and Fluid Machines Lab Total Credits: 1 Contact Hours Practical : 02 Hrs./week, Total Contact Hours 30 hrs Maximum Marks: 50	
Learning Objectives of the Course: 1. Develop a strong understanding of fluid properties, pressure measurement, and flow behaviour through experimentation. 2. Enable students to analyze and interpret experimental data related to fluid flow and hydraulic machinery. 3. Familiarize students with the working and performance evaluation of various hydraulic machines and systems. 4. Bridge the gap between theoretical concepts and real-world applications in fluid mechanics and hydraulics. Course Outcomes (COs) : After completion of the course, students will be able to – 1. Identify and demonstrate the use of various pressure measuring instruments used in fluid mechanics. 2. Evaluate the stability of floating bodies by determining their metacentric height. 3. Perform experiments to determine flow patterns and verify fundamental principles such as Bernoulli's theorem and Reynolds number. 4. Measure flow rates using devices like orifices and venturimeters, and analyze flow losses in pipelines. 5. Analyze the performance characteristics of hydraulic turbines (Pelton, Francis, Kaplan) and pumps (centrifugal, gear). 6. Interpret the working of fluid systems through real-world exposure via industrial visits to water treatment and hydroelectric plants.	
Topics / actual contents of the syllabus	Contact Hours
List of the Experiments 1. Study and performance on different types of pressure measuring devices 2. Determination of metacentric height of a floating body. 3. To perform Reynolds experiment. 4. Verification of Bernoulli's Theorem 5. Measurement of flow by orifice. 6. Measurement of flow by venturimeter. 7. Determination of Major losses in pipe flow 8. Determination of Minor losses in pipe flow 9. Trial on hydraulic turbine-Pelton wheel 10. Trial on hydraulic -turbine Francis 11. Trial on hydraulic- turbine Kaplan turbine 12. Trial on Centrifugal pump. 13. Trial on Gear pump. 14. Visit to hydroelectric power plant. 15. Visit to Water treatment plant	30 Hrs
Students should undertake at least 10 experiments during the semester from above list	

Course Code: MED 253 Course: Strength of materials Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Prerequisite : 1. Fundamentals of Engineering Mechanics. 2. Basic knowledge of material properties. Learning Objectives of the Course: 1. To understand the basic concepts of stress, strain, and their relationship. 2. To learn the behaviour of materials under different loading conditions. 3. To study the concept of bending, shear, and torsion in mechanical components. 4. To analyze columns and struts and determine critical loads. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Explain stress-strain relationship for different materials and loading conditions. 2. Analyze the internal forces in members subjected to axial, shear and bending loads. 3. Determine the stresses in machine components using torsional and bending equations. 4. Evaluate deflection of beams using various methods. 5. Apply concepts of principal stresses and strain energy in engineering problems. 6. Calculate the buckling loads for different types of columns.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Simple Stresses and Strains Stress and strain, Hooke's law, Poisson's ratio, modulus of elasticity, modulus of rigidity, bulk modulus. Stress-strain diagrams for ductile and brittle materials. Bars of varying cross section, composite bars, and thermal stresses. Shear Force and Bending Moment Concept and definition, S.F. and B.M. diagrams for cantilever, simply supported and overhanging beams with point loads, uniformly distributed load, uniformly varying load, and couples. Relationship between load, shear force and bending moment.	10 Hrs
II	Bending and Shear Stresses in Beams Theory of simple bending, bending stress distribution, moment of resistance, section modulus, flitched beams. Shear stress distribution for different sections. Torsion of Circular Shafts Torsion equation, strength and stiffness of solid and hollow circular shafts, torque transmission, combined bending, and torsion.	10 Hrs
III	Principal Stresses and Strain Energy Principal planes and stresses, Mohr's circle theories of failure. Strain energy and resilience under various types of loading.	10 Hrs

	Columns and Struts Euler's and Rankine's formulae for different end conditions, slenderness ratio, critical loads and factor of safety.	
Reference Books: 1. R.K. Bansal, "Strength of Materials", Latest Edition, Laxmi Publications. 2. Gere and Timoshenko, "Mechanics of Materials", Latest Edition, CSB Publishers. 3. S. Ramamrutham, "Strength of Materials", Latest Edition, Dhanpat Rai Publishing. 4. E.P. Popov, "Mechanics of Solids", Latest Edition, Prentice Hall. 5. Timoshenko & Young, "Strength of Materials", Latest Edition, CSB Publishers. 6. S.S. Rattan, "Strength of Materials", Latest Edition, Tata McGraw Hill Publication Co. Ltd.		

Course Code: MED 254
Course: Strength of materials Lab

Total Credits: 1

Contact Hours Practical : 02 Hrs./week, Total Contact Hours 30 hrs

Maximum Marks: 50

Prerequisite:

1. Knowledge of material properties.
2. Basic understanding of mechanics and measurement techniques.

Learning Objectives of the Course:

1. To understand and perform standard material testing experiments.
2. To determine material behavior under tension, compression, bending, torsion and impact.
3. To interpret and analyze test results for engineering applications.
4. To develop skills in using testing equipment and interpreting outcomes.

Course Outcomes (COs) :

After completion of the course, students will be able to –

1. Operate material testing machines and instruments with proper procedure.
2. Perform various mechanical tests like tension, compression, shear, torsion, and impact.
3. Analyze and interpret experimental data.
4. Correlate theoretical knowledge with practical results.
5. Apply testing outcomes in design and material selection.

Topics / actual contents of the syllabus	Contact Hours
List of the Experiments 1. Study and demonstration of Universal Testing Machine. 2. Tension test on mild steel and tor steel. 3. Shear test on mild steel 4. Compression test on timber/wood. 5. Bending test on wooden beam. 6. Torsion test on mild steel. 7. Izod and Charpy Impact test. 8. Brinell and Rockwell Hardness test. 9. Deflection of beams (Simply supported and cantilever beam). 10. Study of strain gauges and measurement of strain in a loaded beam. 11. Determination of modulus of rigidity using torsion apparatus. 12. Study of fatigue testing machine and plotting S-N curve. 13. Assignment on Unit 1. 14. Assignment on Unit 2. 15. Assignment on Unit 3.	30 Hrs
Students should undertake at least 10 experiments during the semester from above list	

Course Code: MED 255 Course: Metrology and Quality Control Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Learning Objectives of the Course: 1. Select appropriate tools and techniques for determining geometry and dimensions. 2. Illustrate the use of Total Quality Control (TQC) tools in quality assurance processes. 3. Design and calibrate measuring tools and equipment accurately for industrial applications. 4. Apply various Quality Control techniques in manufacturing and inspection. 5. Implement Quality Management concepts to enhance productivity and maintain standards. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Recall the basics knowledge of metrology and measuring devices. 2. Explain different linear and angular measuring precise instruments and apply the acquired knowledge for the accurate and precise measurement. 3. Apply knowledge of various tools and techniques used to determine geometry and dimensions of components to produce quality products. 4. Analyze the data of measurement for understanding the concept of quality and Statistical Quality Control. 5. Examine the deviation and surface finish of the measured parts with measuring tools. 6. Discuss the concept of Quality, principles of Statistical Quality Control, seven quality control tools and acceptance sampling.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Metrology Basic Concepts Legal Metrology - Precision - Accuracy - Types of errors - Linear and Angular, Measurements, Standards of Measurements - Slip gauges - Calibration -Interchange ability and selective assembly. Introduction to Comparators Types of Comparators- Mechanical, Mechanical Optical, Electrical and Electronic, pneumatic, Fluid Displacement - Automatic gauging machines. Coordinate Measuring Machine. Introduction to Limits, Fits and Tolerances. Measurement of Internal and External screw threads Measurements of various elements of thread-best size wire-Two and three, wire method.	10 Hrs
II	Surface Finish Measurement and Interferometry Surface Finish Surface topography definitions - Measurement of Surface Texture - Methods - Evaluation of Surface finish. Meaning of RMS and CLA values, Grades of roughness, specifications. Interferometry	10 Hrs

	Principle of light wave interference – Light sources -Types of Interferometers. Measurement of straightness - Flatness- Squareness Parallelism and Circularity.	
III	Statistical Quality Control Introduction - Definition of Quality - Chance Causes and assignable Causes - SQC Benefits and Limitations. Fundamental concepts in probability Normal curve. Theory of Control Charts Control Charts for Variables - X bar and R charts, Standard deviation charts run up - run down - Process capability studies. Control Charts for attributes - Fraction defectives - and number of defects - chart sensitivity. Acceptance Sampling Basic Concepts and OC curve - AQL - LTPD - AOQL - Sampling Plans Simple - Double - Multiple and sequential sampling plans - stratified sampling plans for variables.	10 Hrs
Reference Books: 1. R.K. Jain, "Engineering Metrology", 21st Edition, Khanna Publishers. 2. Hume K.J., "Engineering Metrology", 3rd Edition, Macdonald Publications. 3. R.C. Gupta, "Statistical Quality Control & Quality Management", 9th Edition, Khanna Publishers. 4. Eugene Grant, Richard Leavenworth, "Statistical Quality Control", 7th Edition, Tata McGraw Hill Publication. 5. V.A. Kulkarni, A.K. Bewoor, "Quality Control", 1st Edition, John Wiley Publication. 6. E.O. Doebelin, "Measurement Systems", 6th Edition, McGraw Hill. 7. J.M. Juran, Joseph A. Defeo, "Quality Handbook", 6th Edition, McGraw Hill.		

Course Code: MED 256 Course: Metrology and Quality Control Lab Total Credits: 1 Contact Hours Practical : 02 Hrs./week, Total Contact Hours 30 hrs Maximum Marks: 50	
Prerequisite: 1. Basic Mathematics Learning Objectives of the Course: 1. Selection of tools and techniques for determining geometry and dimensions. 2. Design and calibration of measuring tools and equipment. 3. Application of Quality Control techniques. 4. Application of Quality Management concepts. Course Outcomes (COs) : After completion of the course, students will be able to – 1. Measure linear and angular dimensions using various instruments. 2. Identify and analyze measurement errors and verify component dimensions. 3. Inspect surface quality using optical tools and roughness testers. 4. Interpret component geometry using profile projector, microscope, and CMM. 5. Measure thread and gear parameters using standard metrology tools. 6. Analyze process capability and prepare control charts.	
Topics / actual contents of the syllabus	Contact Hours
List of the Experiments 1. Determination of linear and angular dimensions of a given composite part using precision/non-precision measuring instruments. 2. Error determination with linear/angular measuring instruments. 3. Verification of dimensions and geometry of given components using Mechanical or Pneumatic comparators. 4. Identification of surfaces using optical flat/interferometers and measurement of surface roughness using a surface roughness tester. 5. Determination of geometry and dimensions of a given composite object using a profile projector. 6. Measurement of various angles of a single-point cutting tool using a tool maker's microscope. 7. Measurement of thread parameters using a floating carriage diameter measuring machine. 8. Measurement of spur gear parameters using Gear Tooth Vernier, Span, and Gear Rolling Tester. 9. Determination of given geometry using a Coordinate Measuring Machine (CMM). 10. Determination of process capability from given components and plotting of variable control chart / attribute chart. 11. Assignment on unit 1. 12. Assignment on unit 2. 13. Assignment on unit 3.	30 Hrs
Students should undertake at least 10 experiments during the semester from above list	

Course Code: MDMT-II Course: Robotics, Kinematics and Dynamics (This course will be available for the students from other discipline and Mechanical Engineering students will choose minor course from other discipline) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Prerequisite : 1. Basic Mathematics Learning Objectives of the Course: 1. To provide an introduction to Robotics and Automation including robot classification, design and selection, analysis and applications in industry. 2. To provide information on various types of end effectors, their design, interfacing and selection. 3. To provide the details of operations for a variety of sensory devices that are used on robot, the meaning of sensing, classification of sensor, that measure position, velocity & acceleration of robot joint. 4. To familiarize the basic concepts of transformations performed by robot Course Outcomes (COs) : After completion of the course, students will be able to - 1. Understand basic components of robotics, classification of robots and their applications. 2. Model forward and inverse kinematics of robot manipulators. 3. Analyze forces in links and joints of a robot. 4. Develop a program for robots to perform tasks in industrial applications. 5. Apply basic transformations related to the movement of manipulator. 6. Design a robot mechanism to meet kinematics requirements and to write simple programs.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction: Multibody systems, Automation, Classification of robots, Anatomy, Sensors (velocity, proximity, touch, torque etc), Grippers, selection of Robot based on the Application. Applications: Application of Robots in production systems- Application of robot in welding, machine tools, material handling, and assembly operations parts sorting and parts inspection.	10 Hrs
II	Kinematics: Manipulators Kinematics, Rotation Matrix, Homogenous Transformation Matrix, Direct and Inverse Kinematics for industrial robots for Position and orientation. Demonstrate FK and IK using simulation tools. Robot End Effectors: Introduction, end effectors, interfacing, types of end effectors, grippers and tools.	10 Hrs

	Robotic Sensory devices: Position sensors, Non-optical position sensors - potentiometers, synchros, inductosyn, optical position sensors - opto interrupters, optical encoders.	
III	Differential Kinematics: statics and Dynamics: Jacobian, Lagrangian Formulation, Newton- Euler Formulation for RR & RP Manipulators. Demonstrate differential kinematics using simulation tools. Robot Programming: Robot programming Languages - VAL Programming - Motion Commands, Sensor Commands, End effector commands, and Simple programs.	10 Hrs
Reference Books: 1. R.K. Mittal, I.J. Nagarath, "Robotics and Control", Tata McGraw Hill, 2005. 2. Richard D. Klafter, "Robotic Engineering", Prentice-Hall of India Pvt. Ltd., 2010. 3. Subir Kumar Saha, "Introduction to Robotics", Tata McGraw-Hill Education, 2014. 4. Mark W. Spong, Seth Hutchinson, Mathukumalli Vidyasagar, "Robot Modeling and Control", Wiley, 2006.		

Course Code: VSECT-2
Course: R-Programming

Total Credits: 2

Contact Hours Practical : 04 Hrs./week, Total Contact Hours 60 hrs

Maximum Marks: 50

Prerequisite:

1. Knowledge of any programming method

Learning Objectives of the Course:

1. Understand the R Programming Language.
2. Exposure on visualizing data science problems.
3. Understand the classification and Regression Model.

Course Outcomes (COs) :

After completion of the course, students will be able to –

1. Demonstrate proficiency in R programming syntax, operators, and data structures (vectors, lists, matrices, data frames).
2. Perform data manipulation and export results to specified file locations.
3. Generate basic visualizations such as box plots, scatter plots, histograms, bar charts, and pie charts using R.
4. Create and interpret correlation plots to better understand data relationships.

Topics / actual contents of the syllabus	Contact Hours
List of the Experiments 1. Introduction to R Programming and Study of basic Syntax in R 2. R as a Calculator application: a. Using with and without R objects on console. b. Using mathematical functions on console. c. Write an R script, to create R objects for calculator application and save in a specified location in disk. 3. Descriptive Statistics In R a. Write an R script to find basic descriptive statistics using summary, str, quartile function. b. Write an R script to find subset of dataset by using subset (), aggregate () functions on sample dataset 4. Reading and Writing Different Types of Datasets a. Reading different types of data sets (.txt, .csv) from Web and disk and writing in file in specific disk location. b. Reading Excel data sheet in R. c. Reading XML dataset in R. 5. Visualizations a. Find the data distributions using box and scatter plot. b. Find the outliers using plot. c. Plot the histogram, bar chart and pie chart on sample data 6. Study and implementation of various control structures in R and calculate mean mode median for a dataset. 7. Correlation and Covariance a. Find the correlation matrix. b. Find the outliers using plot. c. Plot the correlation plot on dataset and visualize giving an overview of relationships among data. 8. Regression Model	60 Hrs

Import a data from web storage. Name the dataset and now do Linear/Logistic Regression to find out relation between variables that are affecting the admission of a student in an institute based on his or her entrance score. 9. Classification Model - Install relevant package for classification. - Choose classifier for classification problem. - Evaluate the performance of classifier. 10. Clustering Model - Clustering algorithms for unsupervised classification. - Plot the cluster data using R visualizations. 11. Mini Project	
Students should undertake at least 10 experiments during the semester from above list	

Course Code: AEC-2
Course: Modern Indian Languages

Total Credits: 2

Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs

Maximum Marks: 50

Basket of Modern Indian Languages is available on University Website

Direct link for syllabus:

<http://www.bamu.ac.in/Portals/0/nep-common-AEC-course-sem-I-24-25.pdf>

Or

Visit

<http://www.bamu.ac.in/NEP-Curriculum.aspx#2024>

Entrepreneurship/Economics/Management Courses

Students will have to choose any one theory course form the following Basket. Any

One course in 3rd semester and another course in 4th semester

- 1) Engineering Economics (EEM-I)
- 2) Entrepreneurship Development (EEM-II)
- 3) Industrial Management (EEM-III)

Course Code: EEM-I Course: Engineering Economics (Semester III/ IV) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50 Learning Objectives of the Course: 1. Understand the Time Value of Money 2. Evaluate Engineering Alternatives 3. Analyze Public Sector Projects 4. Understand and Apply Depreciation and Taxation 5. Perform Break-even and Sensitivity Analysis 6. Understand Cost Concepts for Decision-Making Course Outcomes (COs) : After completion of the course, students will be able to - 1. To evaluate Time Value of Money 2. To enable students to perform economic comparisons between different projects or assets 3. To introduce benefit-cost analysis for evaluating the economic feasibility of projects 4. To apply break-even analysis using both linear and non-linear models 5. To develop skills in cost estimation, cost control, cost reduction, and identify relevant costs for decision-making.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Time value of money: Simple and compound interest, Time value equivalence, Compound interest factors, Cash flow diagrams, Calculation, Calculation of time – value equivalences. Present worth comparisons, Comparisons of assets with equal, unequal and infite lives, comparison of deferred investments, Future worth comparison, payback period comparison.	10 Hrs
II	Use and situations for equivalent annual worth comparison, Comparison of assets of equal and unequal life. Rate of return, Internal rate of return, comparison of IIR with other methods, IRR misconceptions. Analysis of	10 Hrs

	public Projects: Benefit/ Cost analysis, quantification of project, cost and benefits, benefit/ cost applications, Cost effectiveness analysis.	
III	Depreciation , Computing depreciation charges, after tax economic comparison, Break-even analysis; linear and non-linear models. Sensitivity analysis: single and multiple parameter sensitivity. Fixed and variable cost , Product and Process Costing, Standard Costing, Cost estimation, Relevant Cost for decision making, Cost estimation, Cost control and Cost reduction Techniques.	10 Hrs

Reference Books:

1. Riggs, J.L., Dedworth, Bedworth, D.B., Randhawa, S.U., "Engineering Economics", Vol. 1, Latest Edition, McGraw Hill International, 1996.
2. James L. Riggs, David D. Bedworth, Sabah U. Randhawa, "Economics for Engineers", Vol. 1, 4th Edition, McGraw-Hill, 2004.
3. Donald Newnan, Ted Eschembach, Jerome Lavelle, "Engineering Economic Analysis", Vol. 1, 8th Edition, Oxford University Press, 2012.
4. John A. White, Kenneth E. Case, David B. Pratt, "Principles of Engineering Economic Analysis", Vol. 1, 6th Edition, John Wiley, 2010.
5. R. Paneerseelvam, "Engineering Economics", Vol. 1, 2nd Edition, PHI, 2008.
6. Michael R. Lindeburg, "Engineering Economics Analysis", Vol. 1, Latest Edition, Professional Publications, 1993.
7. V. Mote, S. Paul, G. Gupta, "Managerial Economics", Vol. 1, Latest Edition, Tata McGraw Hill, 2004.

Course Code: EEM-II Course: Entrepreneurship Development (Semester III/ IV) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Learning Objectives of the Course: 1. Understand the fundamentals of entrepreneurship and its significance in engineering. 2. Identify business opportunities and develop innovative ideas. 3. Create a basic business plan and understand key entrepreneurial strategies. Course Outcomes (COs) : After completion of the course, students will be able to - 1. Explain the concept of entrepreneurship and its importance in the modern economy. 2. Identify and evaluate business opportunities in engineering and technology sectors. 3. Develop a comprehensive business plan including financial, operational, and marketing strategies. 4. Assess the challenges and risks in entrepreneurship and develop strategies to mitigate them. 5. Demonstrate entrepreneurial thinking through case studies, projects, and presentations.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Entrepreneurship Definition, meaning, and characteristics of an entrepreneur, Evolution of entrepreneurship Types of entrepreneurs, Role of entrepreneurship in economic development, Key traits of successful entrepreneurs.	10 Hrs
II	Opportunity Recognition and Idea Generation Identifying business opportunities, Creativity and innovation in entrepreneurship, Techniques for idea generation, Feasibility analysis (technical, market, financial).	10 Hrs
III	Business Planning Components of a business plan, Business models and strategy formulation, Legal requirements for starting a business, Intellectual Property Rights (IPR) and patents.	10 Hrs
Reference Books: 1. C.B. Gupta, Srinivasan, "Entrepreneurship Development", Vol. 1, Latest Edition, Sultan Chand & Sons, 2020. 2. Donald F. Kuratko, "Entrepreneurship: Theory, Process, Practice", Vol. 1, Latest Edition, Cengage, Eric Ries, "The Lean Startup", Vol. 1, Latest Edition, Anonymous, 2011. 3. Alexander Osterwalder, Yves Pigneur, "Business Model Generation", Vol. 1, Latest Edition, Wiley, 2011. 4. Peter F. Drucker, "Innovation and Entrepreneurship", Vol. 1, Latest Edition, Taylor & Francis, 2014.		

Course Code: EEM-III
Course: Industrial management
(Semester III/ IV)

Total Credits: 2

Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs

Maximum Marks: 50

Learning Objectives of the Course:

1. To understand concept of management, administration, Organization, Industrials Laws.

Course Outcomes (COs) :

After completion of the course, students will be able to -

1. Understand the fundamental principles of management.
2. Describe different forms of business organizations and organizational structures.
3. Apply theoretical knowledge to real-world management and organizational challenges.
4. Analyze the impact of organizational, environmental factors.
5. Explain the role of economics in management decision-making.

Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Management: Managing and Manager, organizations and need for management, the managing process, types of managers, the challenge of management, the evolution of Management theory. Management in the 21st century: The importance of organizational and natural environment, elements of direct action environment, managing multiple stock holder relationship, elements of the indirect action environment, Natural Environment management 2000 and beyond, social responsibility and ethics, globalization. Evaluation of case studies related to above concept.	10 Hrs
II	Business Organization Forms of business organization, individual proprietorship, joint stock company, co-operative enterprise, co-operative enterprise and public sector undertakings. Organization structure in industries, Line organization, functional organization, line and staff organization, committee organization, project organization matrix organization. Nature and Significance of Economics. Science, engineering and technology, their relationship with economic development. Basic economic concepts, human wants economic goods, utility value, price cost, wealth and capital. Demand supply, elasticity of demand and supply. Concept of profit and revenues.	10 Hrs
III	Accidents and safety Classification of accidents, according to nature of industries; i.e. fatal, temporary, according to event and place. Causes of accidents , psychological, and other industrial hazards. Effects of accidents. Accident-prone workers, accident to be taken incase of accidents with machines, electric shock, road accident fires and erection and construction accidents	10 Hrs

	Personnel Management: Man power, sources of recruitment, selection and training, job evaluation, performance appraisal, wages and incentives, self and time management.	
Reference Books: 1. James A. F., "Management", Vol. 1, 6th Edition, PHI. 2. Claude S. George, Jr., "Management for Business and Industry", Vol. 1, Revised Edition, Prentice-Hall of India Private Limited. 3. McConnell, Gupta, "Economics: Principles, Problems, and Policies", Vol. 1, 18th Edition, The McGraw-Hill. 4. T.R. Banga, S.C. Sharma, "Industrial Organisation and Engineering Economics", Vol. 1, Latest Edition, Khanna Publishers, Jan-2006. 5. O.P. Khanna, "Industrial Engineering & Management", Vol. 1, Latest Edition, Dhanpat Rai Publication, Jan-2018.		

Value Education Courses (VEC)

Students will have to choose any one theory course form the following Basket. Any
One course in 3rd semester and another course in 4th semester

- 1) Universal Human Values (VECT-I)
- 2) Environmental Students (VECT-II)

Course Code: VECT-I Course: Universal Human Values (Semester III/ IV) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50		
Learning Objectives of the Course: - To appreciate the essential complementarities between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity, which are the core aspirations of all human beings, - To facilitate the development to a holistic perspective among students to lead their Personal and professional lives in an ethical way. - To highlight plausible implications of such a holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour, and mutually enriching interaction with nature. Course Outcomes (COs) : After completion of the course, students will be able to - - Define key terms related to human values. - Explain the concept of happiness as related to right understanding and relationship. - Apply the principles of right understanding in their daily interactions. - Analyze the impact of their values on their behaviour and decisions.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Value Education and Harmony in the Human Being - Understanding Value Education - Self-exploration as the Process for Value Education - Continuous Happiness and Prosperity - the Basic Human Aspirations and their fulfillment - Right Understanding, Relationship and Physical Facility - Happiness and Prosperity - Current Scenario - Method to Fulfill the Basic Human Aspirations - Harmony in the Human Being - Understanding Human being as the Co-existence of the Self and the Body	10 Hrs

	- Distinguishing between the Needs of the Self and the Body - the Body as an instrument of the Self - Understanding Harmony in the Self - Harmony of the Self with the Body - Programme to ensure self-regulation and Health.	
II	Harmony in the Family and Society - Harmony in the Family - the Basic Unit of Human Interaction "Trust" - the Foundational Value in Relationship 'Respect' - as the Right Evaluation Values in Human-to-Human Relationship - Harmony in the Society - Other Feelings, Justice in Human-to-Human Relationship - Understanding Harmony in the Society - Vision for the Universal Human Order	10 Hrs
III	Harmony in the Nature (Existence) and Implications of the Holistic Understanding - Understanding Harmony in the Nature - Interconnectedness, self-regulation, and Mutual Fulfillment among the Four Order of Nature - Realizing Existence as Co-existence at All Levels - The Holistic Perception of Harmony. Implications of the Holistic Understanding - a Look at Professional Ethics - Basis for Universal Human Values - Definitiveness of (Ethical) Human Conduct - Professional Ethics in the light of Right Understanding - A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order - Holistic Technologies, Production Systems and Management Models, Typical Case Studies Strategies for Transition towards Value-based Life and Profession.	10 Hrs

Reference Books:

- P.L. Dhar, R.R. Gaur, "Science and Humanism", Vol. 1, 1st Edition, Commonwealth Publishers.
- Nagaraj, "Jeevan Vidya: Ek Parichaya", Vol. 1, Latest Edition, Jeevan Vidya Prakashan, Amarkantak, 1999.
- A.N. Tripathy, "Human Values", Vol. 1, Latest Edition, New Age International Publishers, 2003.
- E.G. Seebauer, Robert L. Berry, "Fundamentals of Ethics for Scientists & Engineers", Vol. 1, 1st Edition, Oxford University Press.
- M. Govindarajan, S. Natrajan, V.S. Senthil Kumar, "Engineering Ethics and Human Values", Vol. 1, 1st Edition, Prentice Hall of India Ltd.
- B.P. Banerjee, "Foundations of Ethics and Management", Vol. 1, Latest Edition, Excel Books, 2005.
- B.L. Bajpai, "Indian Ethos and Modern Management", Vol. 1, Reprinted Edition, New Royal Book Co., Lucknow, 2008.

E-resources:

- <http://madhyasth-darshan.info/postulations/knowledge/knowledge-of-humane-conduct/>
- https://www.youtube.com/channel/UCQxWr5QB_eZUnwxSwxXEkQw

3. <https://youtu.be/OgdNx0X9231>
4. <https://fdp-si.aicte-india.org/UHV-II%20Practice%20Sessions.php>
5. <https://fdp-si.aicte-india.org/download.php#1/>

Course Code: VECT-II Course: Environmental Studies (Semester III/ IV) Total Credits: 2 Contact Hours Theory: 02 Hrs./week, Total Contact Hours: 30 hrs Maximum Marks: 50 Prerequisites: Understanding of the Concept of Environment Learning Objectives of the Course: 1. To study the environment and ecosystems. 2. To study different types of natural resources. 3. Knowledge and concept of biodiversity and its conservation. 4. Basic knowledge and concept of causes, effects, and control of different types of Environmental pollution. 5. To study population growth and its impact on the environment Course Outcomes (COs) : After completion of the course, students will be able to - 1. Define key environmental terms and concepts (e.g., ecosystem, biodiversity, pollution). 2. Explain the causes and effects of major environmental problems. 3. Apply environmental principles to analyze real-world scenarios. 4. Differentiate between various types of pollution and their impacts.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to environmental studies and natural resources: Definition, scope and Importance and need for public awareness. Natural resources: Forest resources: Use and over-exploitation, deforestation. Timber extraction. Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dam's benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using Mineral resources Food, energy, and land resources: Food resources: world food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems water logging, salinity. Energy resources: Growing energy needs, renewable and non-renewable energy sources, and use of	10 Hrs

	alternate energy sources. Land resources: Land as a resource, land degradation, man-induced landslides, soil erosion, and desertification.	
II	Ecosystems and Biodiversity and its conservation Ecosystems: Concept of an ecosystem. Structure and function of an ecosystem. Producers, consumers, and decomposers. Energy flow in the ecosystem. Ecological succession. Food chains, food webs, and ecological pyramids. Introduction, types, characteristic features, structure, and function of the following ecosystems: Forest ecosystem, Grassland ecosystem, Desert ecosystem, and Aquatic ecosystems (ponds, streams, lakes, rivers, Oceans, estuaries) Biodiversity and its conservation: Introduction Definition: genetic, species, and Ecosystem diversity. Bio geographical classification of India. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity	10 Hrs
III	Environmental Pollution and Social issues and the Environment Environmental Pollution: Definition, Cause, effects, and control measures of Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, nuclear hazards, Role of an individual in the prevention of pollution. Social issues and the Environment: From Unsustainable to sustainable development Urban problems related to energy. Climate change, global warming, acid rain, ozone layer depletion Environment Protection Act. Public awareness.	10 Hrs
Reference Books: 1. Agarwal K.C., "Environmental Biology", Vol. 1, Latest Edition, Nidi Puhl Ltd., Bikaner, 2001. 2. Bharucha Erach, "The Biodiversity of India", Vol. 1, First Edition, Mapin Publishing Pvt. Ltd., Ahmedabad. 3. Heywood V.H., Waston, "Global Biodiversity Assessment", Vol. 1, Latest Edition, Cambridge University Press, 1995. 4. Jadhav H., Bhosale V.M., "Environmental Protection and Laws", Vol. 1, First Edition, Himalaya Publishing House, Delhi. 5. Odum E.P., "Fundamentals of Ecology", Vol. 1, First Edition, W.B. Saunders Co., USA. 6. Miller T.G. Jr., "Environmental Science", Vol. 1, First Edition, Wadsworth Publishing Co.		

Honors Degree Course

Please note that this course is optional and specifically designed for students pursuing a Bachelor's Degree Honors with Multidisciplinary Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option

Course Code: MED 257 Course: Supply Chain Management (Honors) (Semester-IV)		
Total Credits: 3 Contact Hours Theory: 03 Hrs./week, Total Contact Hours 45 hrs Maximum Marks: 100		
Prerequisite : 1. Basic understanding of business and management principles, including functional areas such as marketing, operations, and finance. 2. Fundamental knowledge of operations management concepts such as production planning, process design, and resource utilization. 3. Familiarity with basic quantitative methods or statistics is desirable		
Objectives of the Course: 1. Students will demonstrate fundamental knowledge and structure of supply chain. 2. Students will demonstrate knowledge and understanding of the coordination & usage of IT in supply chain. 3. Students will be able to use the conceptual knowledge to understand other functional aspects of logistic management. 4. Students can apply the various concepts to understand practices of supply chain in various sectors		
Course Outcomes (COs) : After successful completion of the course, students will be able to: 1. Define and explain key concepts, evolution, and strategic challenges in Supply Chain and Logistics Management. 2. Illustrate the structure of Supply Chain including demand, procurement, inventory, and distribution management. 3. Differentiate supply chain design elements such as outsourcing, location decisions, and lean/agile models. 4. Assess the impact of coordination, the bullwhip effect, and IT tools on the performance of supply chains. 5. Demonstrate the application of logistics principles in transportation, packaging, containerization, and material handling. 6. Investigate supply chain practices and strategies across various industries such as agro, Engineering, and Services.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours

I	Basics of Supply Chain Management Key concepts of Supply Chain Management: Concept & Definitions in Supply Chain, Evolution of Supply chain, Participants in supply chain, Strategic challenges, Characteristics of Supply chain. Building structure of Supply Chain Management: Overview of customer focus & demand, Resources & capacity management, Procurement & supplier focus Inventory management, Operations management, Distribution management in SCM. Supply chain design and Planning: Supply chain configuration, Outsourcing and off shoring, Location decisions, Capacity planning, Lean Supply chain, Agile supply chain. Supply chain co-ordination & usage of Information Technology: The bullwhip effect, Coordination in SCM, Collaborative planning, forecasting and replenishment, Data capture and data communications, Data manipulation and reporting, E-business and supply chain integration, new trends in supply chain technology.	15 Hrs
II	Basics of Logistics Management Conceptual framework in logistics: Logistics definition, Scope and function, Logistics management objectives, Customer service aspect in logistics, Supply chain linkages, Role of logistics in supply chain. Transportation: Transportation infrastructure, Freight management, Factors influencing freight cost, Transportation networks, Route planning, Multi modal transportation, Air freight operations and role of shipping. Containerization and Inland Container Depot (ICD): Freight container, Classification of containers, stuffing cargo in containers, Constraints in containerization, ICD functions, Clearance procedure for imports and exports at ICD. Material handling and Packaging: Material handling equipment, automated material handling and ASRS systems, Consumer, and logistical packaging.	15 Hrs
III	Supply Chain in Practice Agro Industry: Food processing, Poultry, Milk products, Fertilizers Engineering Industry: Auto components, Paint industry, Electric Motors Consumer Products Manufacturing Industry: Cosmetics, Mobile Phones, Pharmaceutical Industry, Refrigerator & Air Conditioning Process Industry: Sugar, Cement, Steel, Oil Refineries, Paper Industry (3 Hrs.) Service Industry: Telecom sector, Tourism sector, medical service sector, Food chains, Retail sector	15 Hrs

Reference Books:

1. Michael H. Hugos, "Essentials of Supply Chain Management", 5th Edition, Wiley Publications.
2. Sunil Chopra, Peter Meindl, D. V. Kalra, "Supply Chain Management – Strategy, Planning & Operation", 7th Edition, Pearson Education.
3. Sople, Vinod V., "Logistics Management", 3rd Edition, Pearson Education India.
4. Chandrasekaran, N., "Supply Chain Management: Process, System, And Practice", 1st Edition, Oxford University Press, New Delhi.

Double Minor Course

(This course will be available for the students from other discipline and Mechanical Engineering students will choose double minor course from other discipline)

Please note that this course is optional and specifically designed for students pursuing a Bachelor's Degree with Double Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option.

Course Code: DM-251 Course: Thermal Engineering (Double Minor) (Semester-IV)		
Total Credits: 3 Contact Hours Theory: 03 Hrs./week, Total Contact Hours 45 hrs Maximum Marks: 100		
Prerequisite : 1. Basic knowledge of thermodynamics and fluid mechanics		
Learning Objectives of the Course: 5. To learn about of First law for reacting systems and heating value of fuels. 6. To learn about vapor power cycles and their analysis. 7. To learn about flow of steam through nozzles. 8. To learn the about reciprocating compressors with and without inter cooling 9. To analyze the performance of steam boilers, steam turbines and steam condensers.		
Course Outcomes (COs) : After successful completion of the course, students will be able to: 7. Analyze and evaluate the performance of boilers. 8. Assess the functioning and efficiency of boiler draught systems. 9. Evaluate the performance characteristics of nozzles. 10. Explain the working principles and classifications of condensers. 11. Calculate the efficiency and output of various steam power cycles. 12. Analyze the performance of air compressors.		
Unit No.	Topics / actual contents of the syllabus	Contact Hours
I	STEAM GENERATORS AND PERFORMANCE OF BOILERS Classification, constructional details of process and power boilers, equivalent evaporation, boiler efficiency, energy balance, steam generation controls, introduction to IBR laws, principle and working of high pressure boilers. BOILER DRAUGHT Introduction, classification, determination of height and diameter of chimney, efficiency of chimney, condition for maximum discharge, artificial, forced and induced draught, Advantages.	15 Hrs

II	STEAM NOZZLES Types of nozzles, equation of continuity of nozzle, isentropic flow through nozzle, use of Mollier chart, velocity of steam leaving a nozzle, effect of friction, mass of steam discharged, nozzle efficiency, critical pressure ratio and maximum discharge, supersaturated flow through the nozzle, effect of back pressure on nozzle characteristics. STEAM CONDENSERS Classifications, Jet, Surface and evaporative condensers, comparison, vacuum efficiency, vacuum measurement, mass of circulating water required in a condenser, air removal, capacity of air extraction pumps, introduction to cooling towers.	15 Hrs
III	VAPOUR POWER CYCLES Carnot cycle, ideal Rankine cycle, modified Rankine cycle, Reheat and Regenerative cycles with bleeding of steam, thermal efficiency, specific steam consumption, work ratio, power output, effect of superheat, inlet pressure and back pressure on performance of Rankine cycle. AIR COMPRESSORS: (a) Classifications and working principles, reciprocating compressors. Terminologies used effect of clearance volume, actual indicated diagram, and multistage compression, intercoolers. (b) Rotary compressors, working principles Centrifugal compressor, and axial flow compressor. Comparison between reciprocating and rotary compressors. Vacuum pumps, air motor.	15 Hrs
Reference Books: 1. Nag P. K., "Engineering Thermodynamics", Latest Edition, IMH Publication. 2. Rajput R. K., "A Textbook of Engineering Thermodynamics", Latest Edition, Laxmi Publication. 3. Ballaney P. L., "Thermal Engineering". 4. Domkundwar & Domkundwar, "Introduction to Thermal Power Engineering", Dhanpatrai and Sons, New Delhi. 5. Radhakrishnan, "Fundamentals of Engineering Thermodynamics", PHI.		

